CLASS 10 HOLIDAY HOMEWORK

MATHEMATICS

(Polynomial)

Type 1

1. If $p(x)=3x^3-2x^2+6x-5$, find

i. P(2) ii. P(0)

iii. P(4) iv. P(3)

Type 2

1. Find the value of k, for which one root of the quadratic equation

   $kx^2- 14x+8 =0$ is 2

2. Find the value of k, for which one root of the quadratic equation

   $x^2+8x+k =0$ is -1

3. Find the value of k, for which one root of the quadratic equation

   $x^2+5x+k =0$ is -2

4. Find the value of k, for which one root of the quadratic equation

   $2x^2-11x+k =0$ is 3

Type 3

1. Find the zeros of the following quadratic polynomials and verify the relationship

   between the zeros and the coefficients:

i. $x^2+3x-10$

ii. $x^2+7x+12$

ii. $2x^2-11x+15$

iv. $8.2\sqrt{3}x^2-5x+\sqrt{3}$

v. $x^2+2\sqrt{x}+2$

vi. $8x^2-4$

Type 4

1. Find the zeros of the quadratic polynomial whose

i. Sum = 8 and product = 12

ii. Sum = 2 and product = -6

iii. Sum =5/2 and product= 1

Type 5

1. If a and ß are the zeros of the polynomial $p(x)=x^2-4x+3$. Find

i. $1/\alpha^3 + 1/ß^3$ ii. $\alpha^4ß+\alpha^3ß^4$

iii. $\alpha^3+ ß^3$ iv. $\alpha^3-ß^3$

2. If a and ß are the zeros of the polynomial $p(x)=2x^2+5x+k$ satisfying the relation

$\alpha^2+ß^2+\alpha ß=21/4$. then find the value of k.

(Trigonometry)

1. If cosec x = 2 , find all t- ratio

2. If tan x = 20/21 , show that

   (1 - sin x + cos x)/(1 + sin x + cos x) = 3/7

3. If sinx = show that

   (secx + tanx) = √(b+a)/√(b-a)

4. In a ΔABC, ∠B = 90°, AB = 24 cm and BC =

   7 cm. Find (i) sin A (ii) cos A (iii) sin C (iv) cos C.

5. If cot θ = 7/8, evaluate:

   (i) (1+sinθ)(1−sinθ)/(1+cosθ)(1−cosθ)

6. Evaluate the following:

   a) sin 60. cos30 + cos 60.sin30

   b) 2 $\tan^2$ 45 + cosec^2 30 - $\sin^2$ 60

   c) 2 tan 30/(1 - $\tan^2$ 30)

   d) (1 - $\tan^2$ 45)/(1 + $\tan^2$45)

7. Prove that the following identities

   a) √(1 + sin x)/√(1 - sin x) = sec x + tan x

   b) √(1 + sin x)/√(1 - sin x) = (1 + sin x)/cos x

   c) (cot A - cos A)/(cot A + cos A) = (cosec A - 1)/(cosecA + 1)

   d) (cosec x - sin x)(sec x - cos x) = 1/(tan x + cot x)

   e) (tan x + sec x - 1)/(1 - sec x + tan x) = (1 + sin x)/cos x

8. If $\sec x + \tan x = m$ and $\sec x - \tan x = n$. Prove that $mn = 1$

9. If $\sec x + \tan x = p$. Find

   a) $\sec x$. b) $\tan x$. c) $\sin x$. d) $\cos x$

10. If $(a \cos x + b \sin x) = m$ and

    $(a \sin x - b \cos x) = n$. Prove that $m^2 + n^2 = a^2 + b^2$

SOCIAL SCIENCE

1. "The cement industry is somehow responsible for land degradation."Analyse the statement.

2. "Degradation of land is cause of worry." Explain.

3. "Conservation is resources is vital for development." Explain.

4. Mr. Palani is from Tamil Nadu, wishes to cultivate either tea or wheat. Which one of the crops out of the two can be cultivated in his state?Give your answer with two reasons.

5. In India some reasons are rich in certain types of but deficient in some other type of resources. Do you agree? Support your answers with any three reasons.

Or,

Why should we use natural resources properly and judicially? Explain your views.

6. Why is the issue of sustainability important for development?

7. Analyse the measures adopted to prevent soil erosion which is caused due to natural forces.

8. Explain with examples for accommodative experience of Belgium for peace and harmony.

9. Power sharing has different forms of end roles to play in a society. What are the two main forms of power sharing in modern democracies?

10. Bijay is historian studying the ethnic conflict Sri Lanka. He came across a law passed in 1956 that led significant changes in the country's policies, affecting various communities. Explain three measures taken by Sri Lanka as per the 1956 Act.

11. Evaluate any three prudential and moral reasons for power sharing.

12. " Feeling of alienation was increased among the Sti Lankan Tamils. Which factor is responsible for this.

13. Make a project file on sustainable development on stick file.

ENGLISH

Assign project for class 10th

1) Biography of Robert Frost

2) poetic device

Summer vacation holiday homework

1)-Complete the following chapters

A)- A triumph of surgery

B)- A letter to God

C)-A long walk to freedom

D)- Poetry section

1) Dust of snow

2) Fire and ice

3) write a letter to the Editor of "The Times of India",New Delhi, about water shortage in your colony. You are Rohit Bansal, living in 24, RK Puram, New Delhi.

4) Write the following questions in your own language. (words -100-120 wds)

a. Dr Hariot was a tactful person.Elucidate the statement

b. Trace the character of lencho in your own words (A letter to God).

SCIENCE

(PHYSICS)

Q1) Why do we prefer a convex mirror as a rear - view mirror in vehicles ?

Q2) A concave mirror produces three times magnified (enlarged) real image of an object placed at 10 cm in front of it . Where is the image located?

Q3) Write laws of reflection of light.

Q4) Write uses of concave mirror and convex mirror.

Q5) Define refractive index of a medium.

Q6) Write laws of refraction of light .

Q7) The refractive index of diamond is 2.42. What is the meaning of this statement?

Q8) Define 1 dioptre of power of a lens .

Q9) A convex lens forms a real and inverted image of a needle at a distance of 50 cm from it . Where is the needle placed in front of the convex lens if the image is equal to the size of the object? Also, find the power of the lens .

Q10) Find the power of a concave lens of focal length 2m.

Q11) An object 5 cm in length is held 25 cm away from a converging lens of focal length 10cm . Draw the ray diagram and find the position, size and the nature of the image formed.

Q12) Find the focal length of a lens of power - 2.0 D. What type of lens is this ?

Q13) Project work- Draw formation of images by concave mirror, convex mirror, convex lens and concave lens through ray diagrams in colourful manner on a white chart paper .

(BIOLOGY)

Q.1. Make any two investigatory project from the given topics:-

1. The speed of sprouting seeds.

2. Salt friend or foe.

3. Effect of exercise on blood pressure and Pulse rate.

4. Effects of diet on blood glucose.

Q.2. Complete the intext and text question answers in copy of Ch-Life Processes, Topic - Nutrition in Plants and Animals.

(CHEMISTRY)

1. What is chemical reaction? Write in two or three lines.

2. How slaked lime is formed ? Name the type of reaction?

3. What happens when slaked line reacts with carbon dioxide (Co2). Name the product.

4. Why should a magnesium ribbon be cleaned before burning in air?

5. Write a balanced chemical equation with stating symbols.

solutions of Barium - chloride and sodium sulphate in water react to give insoluble Barium sulphate and the solution of sodium chloride.

6. (a) Define decomposition reaction.

(b) Write one equation each for decomposition reaction when energy is supplied in the form of heat , light and electricity.

7. A solution of substance X is used while washing

(i) Name the substance X and write its formula .

(ii) Write the reaction of the substance X with water.

8. Complete the following reactions and balance it.

(i) Fe + CuSO4 ---> _____ + _____

(ii) Pb(No3)2 -------> (heat) _____+ _____

(iii) AgBr -----> (sunlight) _____+_____

9. Balance the following chemical equations:

NaOH + H2So4 ----------> Na2So4 + H2O

BaCl2+ H2So4-------> BaSo4 + HCl

Zinc + Silver nitrate ---------> zinc nitrate+ silver nitrate

10. Explain the following terms with two examples each:

a. Exothermic reaction. b. Endothermic reaction. c. Oxidation-Reduction Reaction

IT(402)

Unit -1 Digital Document (Advanced)

Chapter 1 – Introduction to Styles

B. Fill in the blanks

1. A _____ is a collection of different formats

2. Styles are especially handy in _____.

3. Proper use of styles improves _____ in a document

4. The first five buttons at the top of the Styles window select the category of _____.

5. Using predefined _____ creates Bookmark in the document.

6. On opening a new file _____Style is used for formatting the document.

7. In page layout documents, you can arrange _____ like text boxes and graphics.

8. Character styles are often integrated _____ Style.

9. _____ allows to apply style at different places in the document.

10. Predefined Style _____be updated by Drag and Drop method.

C. Short answer questions

Q1. What do you understand by styles in LibreOffice writer document?

Q2. Write advantages of using Style over manual formatting, for designing a document.

Ans. Advantages of using Style are :

Q3. What are the different categories of style in LibreOffice writer document?

Q4. Write down the steps to update a style.

Q5. What do you understand by custom styles in LibreOffice writer?

Chapter-2 Working with Images

A. Multiple choice questions

Q1. The text or image which appears faintly in the background of a page is called _____.

(a) Water mark (b) Trade mark (c) Copyright (d) Embossing

Q2. JPG or JPEG in image format stands for _____.

(a) Joint Photographic Experts Group (b) Joint Picture Experts Group

(c) Joint Photographic Experts Graph (d) Joint Photographic Experts General

Q3. In Drawing Object Properties toolbar, Grouping options provided are _____.

(a) Exit Group (b) Ungroup (c) Enter Group (d) All of these

Q4. Which of the following is not the correct file extension for an image file?

(a) GIF (b) JPEG (c) Odt (d) PNG

Q5. Image toolbar does not provide a tool for _____.

(a) filtering (b) cropping (c) copying (d) flipping

C. Short answer type questions

Q1. What is a digital image? How can you create one?

Q2. Write steps to insert an image from the gallery using Drag and Drop method.

Q3. How is resizing of image different from cropping it?

Q4. How is linking of an image different from embedding? Give a situation in which you would prefer to link an image.

Q5. What are the benefits and drawbacks of grouping drawing objects?

Q6. Write factors controlling positioning of an image in a document.

Chapter 3 Advanced Features of Writer

(Section-3, Template)

(Section-4, Table of Contents)

A. State whether the given statements are True or False

1. The topics in Table of Contents are hyperlinked.
2. The Table of Contents in LibreOffice Writer can be updated automatically.
3. TABLE of Contents can be inserted even if the section headings are not styled.
4. Once a ToC is created, it cannot be edited. False
5. We cannot add a graphic as a background of ToC.
6. A single template can be used for multiple documents.
7. A template cannot contain graphics.
8. All documents in Writer are based upon templates.
9. The online templates cannot be added to the list of templates in the templates dialog box.
10. A template once created can be edited again and again.
11. The changes recorded have to be accepted by the original author.
12. We can delete the comments added in a document by the user.

D. Answer the following questions

Q1. What is the need of table of contents?

Q2. What will happen if the ‘Protected Against Manual Changes’ option is not selected in the Type tab of Table of Contents, Index or Bibliography dialog box?

Q3. Define a template.

Q4. Give any one advantage of using a template for your document.

Q5. What is the difference between importing and exporting a template?

Q6. Name any two categories of templates.

Q7. When is exporting of templates useful? Give any one reason.

Q8. What is the difference between Accept Track Change and

Accept All Tracked Changes buttons?

Q9. How do we prepare a document for review?

Q10. Create a project to write the rules to learn Typing. Draw or paste picture of the correct posture while working on computer.

ग्रीष्मावकाश कार्य

(1) कवि सूरदास का चित्र

फाइल पेपर पर बनाएँ और उनकी जीवनी लिखिए|

(2) अर्थालंकार के कितने भेद हैं? नाम ,परिभाषा और उदाहरण सहित फाइल पेपर पर लिखिए|

(3) संकेत बिंदु के अनुसार दिए गए विषय पर 150 शब्दों में निबंध फाइल पेपर पर लिखें|

(i) आत्मनिर्भरता

संकेत -बिंदु:- भूमिका, आत्मनिर्भरता का प्रथम अध्याय, स्वावलंबन की शक्ति, उन्नति का मूल मंत्र, उपसंहार

(ii) सत्संगति

संकेत -बिंदु:- प्रस्तावना, संगति का प्रभाव व अनिवार्यता, सत्संगति से लाभ, उपसंहार

(iii) सड़क सुरक्षा: जीवन रक्षा

संकेत बिंदु:- भूमिका, सड़क सुरक्षा से जुड़े कुछ प्रमुख नियम, सड़क सुरक्षा के नियमों की अनदेखी से होने वाली हानियाँ, इन्हें अपनाने के लाभ, निष्कर्ष

(4) किसी एक विषय पर चार्ट पेपर में चित्र या कोलॉज या न्यूजपेपर कटिंग की मदद से चित्र बनाएँ उनके बारे में लिखें:-

(i)पहलगाम हमला(ii) ऑपरेशन

सिंदूर (iii)मातृ दिवस

(iv) भारतीय सेना की शौर्य शक्ति

(5) किसी एक स्वरचित कविता फाइल पेपर पर लिखें|

या

आप अपनी कोई पसंदीदा कविता फाइल पेपर पर लिखें|

(6) लेखक 'रामवृक्ष बेनीपुरी' का चित्र सहित जीवनी फाइल पेपर में लिखें|